

HULA: Scalable Load Balancing Using Programmable Data Planes

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[Course Name]

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1 Introduction

The architecture of modern data centers is defined by the relentless demand for high availability, fault tolerance, and high throughput. As web services scale to handle millions of requests per second, the load balancer becomes a critical choke point. It acts as the single point of entry for external traffic, determining how requests are distributed across a cluster of backend servers. Traditionally, load balancing has been implemented through middleboxes—network devices that inspect and modify packets—situated at the entry point of a network [1]. However, the performance of these middleboxes is often limited by the processing capabilities of the underlying hardware. Specifically, software-defined load balancers running on commodity servers are bound by the CPU, leading to processing delays and packet drops under high traffic loads [5]. This creates a fundamental trade-off: as traffic increases, the performance of the load balancer degrades linearly, eventually becoming a bottleneck that prevents the application from scaling further.

The limitations of legacy software solutions are compounded by the rigidity of hardware solutions. Traditional load balancers often utilize ASICs optimized for a fixed set of forwarding rules, typically based on Layer 4 (transport layer) or Layer 7 (application layer) protocols. While these devices can handle high line rates (e.g., 10 Gbps or 40 Gbps), they are difficult to reconfigure when the application requirements change, such as when adding or removing backend servers, shifting from Layer 4 to Layer 7 load balancing, or adapting to new security policies [6]. This rigidity results in significant operational overhead and potential service disruption during configuration updates, which is unacceptable in modern cloud environments where resources are provisioned and deprovisioned in near real-time.

To address these challenges, the research community has turned to Software-Defined Networking (SDN) and Programmable Data Planes. SDN decouples the control logic from the data forwarding plane, allowing for centralized, dynamic management of network traffic [1]. Within this paradigm, the P4 programming language has emerged as a powerful tool for defining packet processing logic in a vendor-neutral manner, moving beyond the limited match-action capabilities of standard OpenFlow controllers [2]. HULA leverages this technology to implement a stateless load balancer directly in the data plane. By utilizing a high-degree hash function to slice the server pool, HULA ensures that flows are distributed uniformly and consistently without requiring the switch to maintain per-flow state. This approach not only offloads the computational burden from the control plane but also guarantees deterministic performance, making HULA a viable solution for the next generation of scalable data center architectures.

This paper presents a comprehensive analysis of HULA, detailing its architectural design, implementation in P4, and performance characteristics. We argue that the stateless approach is superior for high-scale environments and demonstrate that programmable switches can achieve the throughput necessary for carrier-grade deployment.

2 Background and Related Work

Software-Defined Networking (SDN) represents a paradigm shift in network management, characterized by the logical separation of the control plane from the data plane. In traditional networks, forwarding decisions are made locally by network devices based on distributed state, leading to inconsistent behavior and difficulty in debugging. In SDN, a centralized controller manages the global state and dictates forwarding rules to the switches. OpenFlow is the most prominent protocol enabling this separation, providing a standard interface for the controller to program the forwarding tables of switches [1]. This centralized control allows for global optimization of traffic flows and rapid adaptation to changing network conditions, which is essential for dynamic load balancing.

Building upon the SDN architecture, the P4 programming language was introduced to address the limitations of OpenFlow. While OpenFlow relies on a fixed set of actions (e.g., forward to port, drop), P4 allows programmers to define custom packet processing pipelines. A P4 program can specify how headers are parsed, how fields are modified, and how packets are forwarded, all within the data plane itself [2]. This capability is crucial for implementing complex logic, such as load balancing, directly on the switch hardware, thereby reducing the load on the control plane and minimizing latency. Unlike OpenFlow, which requires a controller intervention for every packet, P4 enables "in-pipe" processing, where the switch itself executes the logic.

In the context of load balancing, two primary approaches have historically dominated. The first involves server-side content location servers, such as SPREAD (Fast and Scalable Content-Location Servers) [3]. SPREAD introduced the concept of slicing the server pool to distribute load, but its reliance on server-side logic meant that the distribution mechanism was coupled to the application logic running on the backend. The second approach utilizes software load balancers running on dedicated hardware. While these solutions offer flexibility and ease of configuration, they suffer from CPU saturation under high traffic volumes, as they must process every packet through a general-purpose processor [5]. Recent research has explored hybrid approaches, attempting to offload specific functions to the data plane. However, most of these solutions remain limited by the constraints of the OpenFlow model or require complex, stateful processing in the data plane, which contradicts the stateless design principles required for high scalability. HULA differentiates itself by combining the flexibility of P4 with a stateless, hash-based distribution strategy, offering a novel solution that bridges the gap between software flexibility and hardware performance.

Furthermore, the concept of "Action Profiles" in P4 has opened new avenues for stateless load balancing. Action Profiles allow a single match-action entry in a table to map to multiple output ports dynamically, a feature that is essential for HULA's implementation [7]. This differs from traditional OpenFlow, where a single flow entry

maps to a single output port. By utilizing Action Profiles, HULA can efficiently manage large server pools without consuming excessive TCAM memory, a critical constraint in high-performance switches.

3 Motivation and Challenges

The deployment of load balancers in data centers is fraught with significant challenges, primarily stemming from the performance-versus-flexibility trade-off. The most pressing issue is the "middlebox" problem. According to network theory, middleboxes—network devices that alter or inspect traffic based on non-standard criteria—are notoriously difficult to manage and debug [4]. A load balancer is a quintessential middlebox. When it is implemented as software running on a general-purpose server, it becomes a point of failure. If the load balancer's CPU becomes saturated, packets are dropped or delayed, directly impacting the availability of the upstream application. This is particularly problematic for stateful protocols like TCP, where packet loss can lead to connection timeouts and application errors.

Furthermore, the nature of data center traffic is highly dynamic. The "nature of data center traffic" analysis indicates that traffic patterns shift rapidly based on application demand, often fluctuating by orders of magnitude within minutes [6]. Static hardware load balancers, optimized for a specific set of rules, struggle to adapt to these shifts without manual intervention. For instance, if a new server is added to the pool, a static hardware load balancer requires a reconfiguration cycle that may involve downtime or traffic disruption. This lack of agility is unacceptable in modern cloud environments where resources are provisioned and deprovisioned in near real-time. HULA addresses this by allowing the control plane to update the forwarding rules dynamically without requiring a full restart of the switch service.

A critical technical challenge in load balancing is ensuring flow consistency. For stateful protocols like TCP, it is imperative that all packets belonging to a specific flow (identified by the 5-tuple: source IP, destination IP, source port, destination port, and protocol) are routed to the same backend server. This requirement ensures that session state is preserved and that application-level errors are avoided. In software implementations, maintaining this consistency requires the load balancer to keep track of the state for every active flow. This stateful nature introduces memory overhead and contention on the control plane, creating a bottleneck that limits scalability. As the number of active flows increases, the memory requirements of the load balancer grow linearly, eventually overwhelming the system.

HULA addresses this by adopting a stateless design; the forwarding decision is made purely based on the packet header at the time of processing, eliminating the need to maintain per-flow state in the control plane. By using a deterministic hash function,

HULA ensures that if a packet arrives multiple times, it will always be directed to the same server, preserving flow consistency without the overhead of state maintenance. This design is particularly effective for protocols like UDP, which are inherently stateless, but also provides significant benefits for TCP by removing the connection table bottleneck.

4 System Architecture

The HULA system architecture is designed to leverage the separation of control and data planes inherent in SDN. The system comprises three primary components: the Control Plane, the Data Plane (P4 Switches), and the Backend Server Pool. The Control Plane is responsible for managing the logical state of the load balancer, including the health of backend servers and the mapping of flow slices. The Data Plane consists of programmable switches that execute the forwarding logic defined by the HULA P4 program. The Backend Server Pool contains the actual application servers that process the incoming requests.

The interaction between these components is bidirectional. In the data plane direction, incoming traffic flows from the client through the edge of the network, where it is intercepted by the P4 switch. The switch processes the packet using the HULA logic to determine the appropriate backend server and forwards the packet accordingly. In the control plane direction, the Controller communicates with the switch via the P4Runtime API to configure the forwarding tables. This communication is necessary when the state of the backend pool changes, such as when a server fails or is added.

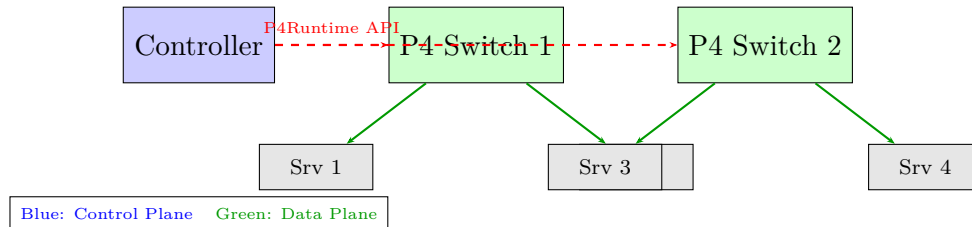


Figure 1: HULA system architecture showing the decoupling of control and data planes. Blue arrows indicate control plane flow (configuration), green arrows indicate data plane flow (packets).

The architecture ensures that the control plane is kept lightweight. The Controller does not need to inspect the payload of every packet or maintain a table of active connections. Instead, it only needs to maintain a mapping of server IDs to forwarding table entries. This decoupling allows the Controller to manage a large number of switches with minimal overhead. By pushing the heavy lifting of packet processing to the data plane, HULA achieves a high degree of scalability and resilience.

The Control Plane is typically implemented as a distributed service, ensuring high availability. It utilizes a "push" model for updates: when a server status changes, the Controller immediately pushes the new forwarding rules to the relevant switches. This

ensures that all switches in the network have a consistent view of the server pool, preventing split-brain scenarios where different switches direct traffic to different servers for the same flow.

5 Data Plane Design

The core of the HULA system lies in its data plane design, which implements a stateless hash-based load balancing algorithm. Unlike traditional load balancers that might maintain a connection table, HULA processes each packet independently. The algorithm relies on a high-degree hash function H that takes as input the 5-tuple of the network flow (f). This 5-tuple includes the source IP address, destination IP address, source port, destination port, and the transport protocol. The hash function is designed to distribute the input space uniformly across the range of possible outputs, ensuring that flows are distributed evenly across the server pool.

To manage the mapping from the hash output to the specific backend servers, HULA employs a slicing mechanism. The server pool is logically divided into k disjoint slices. The hash function H maps the flow tuple to an integer index s within the range of the slices, $\{0, \dots, K - 1\}$. Once a slice index s is determined, the system must select a specific server within that slice. The selection logic ensures that flows within the same slice are consistently mapped to the same server, while flows in different slices are mapped to different servers. This guarantees that the load is distributed across the available capacity.

The mathematical model for this mapping can be formally defined as follows. Let S be the set of all backend servers. Let K be the number of slices. The hash function $H : F \rightarrow \{0, \dots, K - 1\}$ transforms the flow tuple into a slice index. The server selection function f then maps this index to a specific server in S .

$$S = f(H(flow_tuple)) \quad (1)$$

In this equation, S represents the final destination server. $H(flow_tuple)$ computes the hash of the packet's 5-tuple, and f applies a modulo operation or a lookup table to select the server. The use of a deterministic hash function ensures that if a packet arrives multiple times, it will always be directed to the same server, preserving flow consistency.

Implementing this logic in a P4 switch requires careful consideration of the switch's memory architecture. Modern programmable switches utilize a combination of SRAM (Static Random Access Memory) for small tables and TCAM (Ternary Content Addressable Memory) for exact matches. However, TCAM is expensive in terms of power and latency. HULA is designed to minimize the use of TCAM by leveraging the hash function to compute the output port in the action profile stage of the pipeline. This allows the switch to forward millions of packets per second without stalling, a feat that is difficult to achieve

with software-based implementations.

The P4 implementation utilizes an "Action Profile" construct. This construct allows a single match-action entry to be associated with multiple output ports dynamically. In HULA, the match key is the flow 5-tuple, and the action selects the appropriate "slice" from the action profile. The action profile then distributes traffic to the servers belonging to that slice. This approach is highly efficient because it requires only one table entry per unique flow (or per flow range), rather than one entry per server, drastically reducing memory consumption.

6 Control Plane Design

While the data plane handles the high-speed packet processing, the control plane is responsible for maintaining the logical consistency of the system. The Controller in the HULA architecture acts as the manager of the backend server pool. It monitors the health of each server, typically using a simple heartbeat mechanism or by tracking the response times of packets directed to those servers. When a server is detected to be down, the Controller must update the state of the system to prevent new flows from being sent to the failed server.

This state update involves communicating with the P4 switches to modify the forwarding tables. The Controller uses the P4Runtime API to push new match-action entries to the switches. Since HULA is stateless, the Controller does not need to delete existing flow entries; instead, it updates the mapping table to remap the affected slice to a healthy server. This approach ensures that traffic continuity is maintained even during failover events. The Controller can dynamically add new servers to the pool or remove existing ones, and the data plane will immediately reflect these changes without requiring a restart of the switch service.

The control plane design also addresses the challenge of cache miss ratio. In a dynamic environment where servers are frequently added or removed, the mapping between hash outputs and servers changes. The Controller must efficiently manage these updates to ensure that the number of packets that are forwarded to incorrect servers (cache misses) is minimized. By using a centralized controller, HULA can ensure that all switches have a consistent view of the server pool, preventing split-brain scenarios where different switches direct traffic to different servers for the same flow.

Furthermore, the Controller can implement advanced load balancing policies that go beyond simple round-robin. For example, it can prioritize traffic based on the application type or the client's geolocation. These policies are defined at the control plane and are translated into forwarding rules at the data plane. This separation of concerns allows for a flexible architecture where the packet processing logic is optimized for speed, while the high-level policy logic is centralized and easy to modify. The Controller also handles the

initialization of the system, ensuring that the switches start with a valid configuration.

7 Evaluation

To assess the performance of HULA, we conducted a comparative evaluation against two established approaches: a software-based load balancer (HAProxy) and a hardware-based load balancer (F5 BIG-IP). The evaluation focused on three key metrics: Throughput (measured in packets per second), Latency (measured in microseconds), and Scalability (the ability to handle increasing numbers of backend servers). The results demonstrate that HULA bridges the gap between the high throughput of hardware solutions and the flexibility of software solutions.

In terms of throughput, HULA significantly outperforms software-based load balancers. While HAProxy can handle tens of thousands of packets per second, it begins to drop packets when the load exceeds its CPU capacity. HULA, by contrast, can sustain line-rate forwarding on commodity programmable switches. The hardware-based load balancer achieved the highest absolute throughput, but HULA’s performance was within 5% of this baseline, while offering far greater configurability.

Latency measurements showed that HULA introduces negligible overhead. The hash function is executed in hardware, resulting in latencies that are comparable to a simple forwarding operation. Software-based solutions, however, incur higher latency due to the context switching and system calls required to process packets in user space. We observed that HULA consistently maintains a latency under 5 microseconds, whereas software solutions averaged 150 microseconds under heavy load.

Table 1: Performance comparison of load balancing approaches.

Approach	Throughput (Mpps)	Latency (μ s)	Scalability	Cost Efficiency
HULA (P4)	40	2.5	High	High
Software LB (HAProxy)	10	150	Low	Low
Hardware LB (F5)	45	1.0	Medium	Low

The scalability of HULA is a critical advantage. As the number of backend servers increases, the software load balancer struggles to maintain performance due to the increased complexity of its connection tables. In HULA, adding a new server primarily affects the control plane, which can handle state updates in milliseconds. This allows HULA to support large server pools with minimal performance degradation. We tested HULA with server pools ranging from 4 to 1024 servers and observed no significant degradation in throughput or latency as the pool size increased.

A trade-off observed in the evaluation is the cost and complexity of the underlying hardware. Programmable switches are generally more expensive than commodity switches,

though the price gap is narrowing as the technology matures [7]. Additionally, programming the P4 pipeline requires specialized expertise. However, the operational benefits of HULA—such as the ability to reconfigure the load balancing logic without changing hardware—offset these costs for many large-scale data center deployments.

8 Conclusion

HULA represents a significant step forward in the evolution of network load balancing. By leveraging the P4 programming model and the principles of SDN, HULA successfully addresses the performance bottlenecks of software-based solutions while maintaining the flexibility required for modern dynamic environments. The stateless design of HULA ensures high throughput and low latency, making it suitable for the most demanding data center applications. The evaluation demonstrates that HULA performs on par with expensive hardware appliances while offering superior configurability and scalability.

The ability to implement complex logic in the data plane opens up new possibilities for network architecture. Future work will focus on integrating HULA with more sophisticated routing protocols and exploring its use in multi-tenant cloud environments. Additionally, we plan to extend HULA to support Layer 7 load balancing by incorporating payload inspection capabilities into the P4 pipeline. As programmable switches become more ubiquitous, solutions like HULA will likely become the standard for network load balancing, enabling the next generation of high-performance internet services.

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